

**Ismail AISSAOUI**

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To the PhD Selection Committee  
**EDT Program - Catalyst Project**

Attn: Prof. Fabrice Gamboa, Dr. Mathieu Vallee, Dr. Clément Gauchy

April 30, 2026

**Subject: Application for PhD: Interpretable and Robust Machine Learning for Physics-Based Numerical Simulations (Rank 1 - PC1)**

Dear Members of the Selection Committee,

As a State Engineer in Artificial Intelligence from ESI-SBA, I am writing to express my enthusiastic application for the PhD position in "Interpretable and Robust Machine Learning for Physics-Based Numerical Simulations" within the Engineering Digital Twin (EDT) national program. This application is motivated by the close alignment between my technical background in hybrid physics-data modeling and the research objectives of the Catalyst project.

Currently, I am completing my thesis at Sonatrach, where I have developed a **Physics-Informed Digital Twin** for gas turbine fleet monitoring. My research focuses on using machine learning to emulate and accelerate numerical solvers for complex physical problems—a direct parallel to the objectives of the **Catalyst (Reliable Hybrid Model Forge)** project. Specifically, I have designed custom **PINN (Physics-Informed Neural Network)** loss functions that enforce thermodynamic consistency within generative surrogate models based on **Mamba-SSM** and **xLSTM** architectures.

The core challenge of your project—addressing the "black-box" nature of ML surrogates and their sensitivity to out-of-distribution data—is exactly what motivates my current work. In my Sonatrach project, I have encountered the critical need for **quantitative indicators of model reliability** when deploying surrogates on edge devices. I am particularly eager to apply advanced statistical methods, such as **Sensitivity Analysis**, **Optimal Transport**, and **Statistical Depth**, to define the reliability domains of these hybrid models.

My background in high-dimensional statistical analysis and my experience with production-grade PyTorch and C++ implementations provide me with the necessary tools to contribute to the **Artemis platform** and the broader EDT ecosystem. I am a highly autonomous researcher, comfortable bridging the gap between theoretical statistics and industrial physical simulations.

Thank you for your time and consideration. I look forward to the possibility of discussing how my experience in physics-aware AI and industrial surrogates can contribute to the success of the Catalyst project.

Sincerely,

**Ismail AISSAOUI**